Jerry Peng

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Education

Carnegie Mellon University | College of Engineering — **Expected Graduation: December 2026**

M.S. in Mechanical Engineering | Concentration Area: Robotic & Control Systems | GPA 4.0/4.0

Relevant Coursework: Electro-Mechanical Systems Design, Modern Control Theory, Optimal Control & Reinforcement Learning, Non-Linear Control, Robot Dynamics, ML/AI for Engineers

University of Pittsburgh | Swanson School of Engineering — **May 2025**

B.S. in Mechanical Engineering; Honors Degree | GPA 3.9/4.0

Relevant Coursework: Mechatronics, Robotics, Automatic Controls, Dynamic Systems

Relevant Experience

Boat Project | Robomechanics Lab — **Pittsburgh, PA**

Graduate Research Assistant — **September 2025 – Present**

- Developing a controller to devise optimal steering configurations for unmanned surface vessels (USVs)
- Simulating control algorithm outputs using Unreal Engine 5 and Open Sound Control to validate effectiveness

Professor Bajaj's Research Group — **Pittsburgh, PA**

Undergraduate Research Assistant — **August 2024 – May 2025**

- Devised a prototype 3-D Printer to create Core-Shell-Shell filaments for the fabrication of new dielectric elastomer actuators
- Utilized COMSOL to simulate fluid flow and determine rheometry of liquid within the 3-D Printer for a variety of designs

ESTAT Actuation — **Pittsburgh, PA**

Mechanical Engineering Co-op & Rotary Lead — **January 2024 – August 2024**

- Coordinated with billion-dollar clients to ensure product quality and consistency, helping generate over $100,000 dollars in revenue
- Refined the development process of new custom rotary clutch designs by reducing process time by over 50% and saving over 100+ hours in product development
- Designed over 20 custom rotary clutch designs, culminating in a product catalog featured at the 2024 Automate Show

MEMS 0024: Introduction to Mechanical Design — **Pittsburgh, PA**

Undergraduate Teaching Assistant (TA) — **August 2023 – December 2024**

- Demonstrated a high level of proficiency in Solidworks Modeling, Simulation and Engineering drawings through running weekly recitations and office hours

Academic Projects

Launch-a-Birdie | Carnegie Mellon University — **August 2025 – December 2025**

- Engineered a programmable shuttlecock launcher prototype with configurable aiming and launch controls

Senior Design Project | University of Pittsburgh — **August 2024 - December 2024**

Automation of Joule-Annealing of Amorphous, Nano-crystalline Soft Magnetic Materials

- Implemented a codebase to automate live plotting and data collection of the Joule Annealing process for Soft Magnetic Ribbons
- Reduced production time by 300% for each test performed and incorporated in process monitoring of resistivity to optimize material properties and allow for increased rates of design iteration

Leadership Experience

President | Asian Christian Fellowship | University of Pittsburgh — May 2024 - May 2025

Secretary | Badminton Club at Pitt | University of Pittsburgh — January 2024 - May 2025

Skills

Design: CAD (Solidworks), Simulation (ANSYS, Solidworks, COMSOL, UE5), PCB Design (Altium Designer, KiCAD, EasyEDA, Fusion 360), DXF Creation (Adobe Illustrator, CorelDRAW), Microsoft Office Suite (Word, Excel, PowerPoint)

Control: PID, LQR, MPC, Microcontroller Communication (SPI, I2C, USART), Motor Control, Simulink

Technical Expertise: Parts Sourcing, FMEA, DFM, Stress/Structural Analysis, Data Modeling, Quality Control, Engineering Documentation, Tolerance Analysis, GD&T, Linux

Programming: MATLAB, Python, C++ (Arduino IDE), C, ROS, Java

Prototyping: Laser Cutter (ULS PLS6.75), Waterjet (Wazer), Lathe, Mill, 3D-Printing (SLA & FDM), Chop Saw, Band Saw, Drill Press, Soldering, Hand Tools